# Fine-Tuning DistilGPT-2 for Sentiment Classification and Measuring Catastrophic Forgetting on a Boolean QA Task

**Final Project Report**
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## 1: Introduction and Project Statement

### 1.1 Background

Large language models (LLMs) have transformed natural language processing by demonstrating strong performance across a wide range of tasks such as text classification, question answering, summarization, and text generation. Despite these capabilities, fine-tuning LLMs on specific downstream objectives often leads to a well-known issue called catastrophic forgetting. This phenomenon occurs when a model, after being trained on a new task, loses its ability to perform previously learned tasks with the same effectiveness. The issue has been discussed in foundational work by McCloskey and Cohen in 1989 and further examined by French in 1999. Catastrophic forgetting presents a significant challenge in realistic machine learning workflows because many organizations depend on adapting pre-trained models to narrow domains while still expecting them to retain broad generalization abilities. Understanding how forgetting arises, how severe it is, and what factors contribute to it is essential for developing strategies that preserve overall model competence.

### 1.2 Project Statement

This project investigates catastrophic forgetting in transformer-based language models by examining how fine-tuning DistilGPT-2 for sentiment classification affects its ability to perform Boolean question answering. The approach involves fine-tuning the model on a multiclass sentiment classification dataset using an instruction-style prompting format, then evaluating how this fine-tuning changes the model's performance on the BoolQ dataset. Forgetting is quantified by comparing the model's accuracy on BoolQ before and after fine-tuning. The project aims to analyze the extent of the trade-off between improvements on the fine-tuned task and degradation on an unrelated secondary task, providing insight into how narrow task optimization impacts general reasoning abilities.

### 1.3 Research Questions

This study centers on three main questions. The first question explores how much performance improvement DistilGPT-2 achieves on sentiment classification after fine-tuning. The second examines whether the fine-tuning process causes measurable degradation on an unrelated reasoning task such as BoolQ. The third investigates the magnitude of the trade-offs involved, specifically how gains in sentiment classification accuracy correspond to losses in Boolean question answering performance. These questions guide the evaluation of the model's behavior and help reveal the broader implications of task-specific fine-tuning for compact language models.

### 1.4 Significance

The results of this project contribute to ongoing research in transfer learning and continual learning within natural language processing. Understanding how and why catastrophic forgetting occurs is important for anyone deploying fine-tuned language models in real applications. The findings can inform deployment strategies, highlight the need for forgetting-mitigation techniques, and shed light on the capacity limitations of smaller transformer models such as DistilGPT-2. By examining forgetting through a controlled experimental setup, this work provides a clearer picture of the trade-offs involved when specializing a general-purpose language model for a focused task.

## 2: Data Sources

### 2.1 Sentiment Classification Dataset

The sentiment classification dataset used in this project comes from the Hugging Face Hub, specifically the `Sp1786/multiclass-sentiment-analysis-dataset` repository. It contains a large collection of short text snippets and social media posts annotated with one of three sentiment categories: negative, neutral, or positive. The dataset is divided into three splits, consisting of 31,232 training examples, 5,205 validation examples, and 5,206 test examples. Each label corresponds to a different type of emotional expression within the text. Negative examples typically reflect dissatisfaction, complaints, or critical statements. Neutral examples contain objective descriptions or emotionally ambiguous content. Positive examples reflect praise, satisfaction, or other upbeat emotional tones. Sample entries illustrate the variety of the dataset, with texts such as “Cooking microwave pizzas, yummy,” labeled as positive, “Any plans of allowing sub tasks to show up in the widget?” labeled as neutral, and “This is the worst experience I've ever had,” labeled as negative. The dataset is well suited for evaluating how effectively fine-tuning can help a language model learn sentiment-related textual patterns.

### 2.2 Boolean Question Answering (BoolQ) Dataset

The secondary dataset used for measuring catastrophic forgetting is the BoolQ dataset, originally introduced by Clark and colleagues in 2019 and available on the Hugging Face Hub under `google/boolq`. BoolQ is a reading comprehension dataset in which each example consists of a short passage from Wikipedia, paired with a yes/no question that can be answered solely from the information contained in the passage. Unlike sentiment classification, BoolQ requires the model to engage in passage-grounded reasoning rather than emotional interpretation. For this project, a balanced evaluation subset of 500 examples was sampled from the dataset's validation split, containing roughly equal proportions of "yes" and "no" answers. This dataset was chosen specifically because its task structure differs substantially from sentiment classification, making it an effective benchmark for detecting catastrophic forgetting. Since the two tasks rely on different forms of textual understanding, any degradation in BoolQ performance after fine-tuning provides strong evidence that the model has overwritten or altered internal representations necessary for its original reasoning ability.

The contrast between these two datasets allows the project to examine how task-specific fine-tuning affects a model's ability to generalize across domains. The sentiment dataset tests emotional and stylistic understanding of short texts, while BoolQ evaluates factual reasoning grounded in contextual passages. By pairing them, the study can clearly quantify whether training on one form of textual interpretation compromises another.

# 3: Technologies Used

## 3.1 Core Framework

The implementation of this project relied on several core software frameworks widely used in modern machine learning research. Python 3.10 served as the primary programming language, given its extensive ecosystem of scientific computing tools. PyTorch 2.0 provided the foundational deep learning framework used to load, train, and run the DistilGPT-2 model. The Hugging Face Transformers library, version 4.35, enabled streamlined access to the model architecture, tokenizers, training utilities, and configuration options necessary for fine-tuning. The Datasets library, version 2.14, supported efficient loading, preprocessing, and management of both the sentiment classification dataset and the BoolQ evaluation set. These tools together formed the backbone of the model training and evaluation pipeline.

## 3.2 Model Architecture: DistilGPT-2

DistilGPT-2 was chosen as the base model because it offers a useful balance between computational efficiency and expressive capacity. The architecture is a distilled version of GPT-2, retaining the decoder-only transformer structure while reducing the total number of parameters to approximately eighty-two million. The model supports a vocabulary of 50,257 tokens and accepts sequences up to a context length of 1,024 tokens, although this project truncated inputs to a maximum of 128 tokens to reduce memory usage and training time. Since DistilGPT-2 is created through knowledge distillation, it inherits many of the capabilities of the original GPT-2 model while operating with fewer parameters and faster inference speed. These characteristics make it an ideal choice for experiments involving task specialization and forgetting, as the model is small enough to fine-tune efficiently while still capable of learning meaningful linguistic representations.

### 3.3 Analysis and Visualization Libraries

The evaluation and visualization components of the project were built using several standard scientific Python libraries. NumPy supported numerical computations and data manipulation throughout the analysis. Matplotlib was used to generate high-quality plots, including accuracy comparison charts and summary figures illustrating forgetting. Seaborn provided additional statistical visualization functions, especially useful for producing clear and aesthetically consistent heatmaps and distributions. The scikit-learn library facilitated the computation of evaluation metrics such as confusion matrices, enabling a deeper analysis of class-by-class performance on the sentiment task. Progress indicators during dataset processing and evaluation were handled with tqdm, which allowed for real-time monitoring of batch operations.

### 3.4 Development Environment

The model training process was carried out using Google Colab, which provided access to GPU acceleration through hardware such as NVIDIA T4 or V100 GPUs. This environment allowed the fine-tuning of DistilGPT-2 within reasonable computational limits, given the model's size and the number of training examples involved. All evaluation experiments were executed on a local machine using CPU resources, since inference on the fine-tuned model required relatively modest computational power. Git and GitHub were used for version control, ensuring that code, notebooks, results, and documentation were organized and tracked throughout the project's lifecycle. To support reproducibility, all experiments used a consistent random seed of forty-two across Python, NumPy, and PyTorch.

## 4: Methods Employed

### 4.1 Experimental Pipeline

The experimental workflow for this project followed a structured three-stage pipeline implemented across three separate Jupyter notebooks. The first stage focused on preparing both datasets. The sentiment dataset was loaded from Hugging Face, converted into an instruction-style prompt format, and tokenized using the DistilGPT-2 tokenizer with a maximum input length of one hundred twenty-eight tokens. The processed training and test splits were then saved for later use. During this stage, a balanced evaluation subset of five hundred examples was also constructed from the BoolQ validation set in order to create a standardized baseline for measuring performance before and after fine-tuning. The second stage centered on fine-tuning and evaluation. The pretrained DistilGPT-2 model was first evaluated on both sentiment classification and Boolean question answering to establish baseline performance levels. The model was then fine-tuned on the sentiment classification dataset using the instruction-style prompts. After training, the model was re-evaluated on both tasks so that any changes in performance could be directly measured. The third stage involved analyzing and visualizing results. Accuracy metrics were compiled, comparison charts were created, confusion matrices were generated for detailed sentiment evaluation, and qualitative examples were examined to understand how the model's behavior changed after fine-tuning.

### 4.2 Instruction-Style Prompt Engineering

To reformulate sentiment classification as a text generation problem, each example in the dataset was transformed into an instruction-style prompt. This format presented the input text followed by a question asking for the sentiment, along with the expected answer. The structure encouraged the model to treat the classification process as a natural language generation task rather than a traditional supervised classification problem. During training, the model learned to complete these prompts by generating the appropriate sentiment label. This approach aligns with recent trends in instruction tuning, in which models are guided to interpret inputs and produce outputs within a conversational or question-answering framework.

### 4.3 Training Configuration

The fine-tuning process used a compact yet effective training configuration designed to balance compute efficiency and performance. The model was trained for two epochs with a batch size of two, and gradient accumulation was applied across eight steps to achieve an effective batch size of sixteen. The learning rate was set to five times ten to the negative fifth, with a weight decay value of zero point zero one applied to regularize the training process. The AdamW optimizer was used because of its stability and its widespread adoption in fine-tuning transformer-based models. This configuration allowed the model to converge efficiently while minimizing the risk of overfitting to the sentiment dataset.

### 4.4 Forgetting Metric

To quantify catastrophic forgetting, this project compared the model's performance on the BoolQ dataset before and after fine-tuning. The forgetting metric was defined as the difference between baseline BoolQ accuracy and post-fine-tuning BoolQ accuracy. A positive value indicated that the model's performance on BoolQ had degraded as a result of fine-tuning. Because BoolQ and sentiment classification involve distinct cognitive skills within the model, changes in BoolQ performance after training served as a direct measure of whether the fine-tuning process overwrote or altered representations that were previously useful for reasoning over passages.

## 5: Results

### 5.1 Quantitative Results Summary

The evaluation results reveal clear changes in the model's performance before and after fine-tuning. Prior to any task-specific training, the baseline DistilGPT-2 model achieved an accuracy of 59.00 percent on the sentiment classification test set. After fine-tuning, its accuracy on this task increased substantially to 72.00 percent, demonstrating that the model effectively learned sentiment-specific patterns from the instruction-style training examples. In contrast, the model's performance on the BoolQ dataset declined measurably following fine-tuning. The baseline BoolQ accuracy of 57.80 percent dropped to 50.40 percent, a decrease of 7.40 percentage points. This reduction places the model's post-training BoolQ performance almost exactly at the level of random guessing, since BoolQ is a binary yes-or-no task. The difference between the baseline and fine-tuned BoolQ accuracies serves as the model's forgetting score, which in this case is 7.40 percent. Taken together, these results indicate a substantial improvement on the target task accompanied by meaningful degradation on the secondary task.

## 5.2 Sentiment Classification Analysis

The improvement in sentiment classification performance reflects the model's ability to learn emotion-related linguistic patterns once exposed to targeted examples. A confusion matrix constructed from a representative sample of one hundred test cases helps illustrate the model's specific strengths and weaknesses after fine-tuning. Within this sample, the model correctly identified twenty-seven of thirty-seven negative instances, twenty-three of thirty-four neutral instances, and twenty-two of twenty-nine positive instances. The misclassifications tended to occur between adjacent classes, particularly between negative and neutral sentiments or between neutral and positive sentiments. Extreme misclassifications, such as confusing highly negative statements with highly positive ones, occurred rarely. These results suggest that the model formed a reasonably well-calibrated internal representation of sentiment categories, although it still struggled with borderline or ambiguous cases.

## 5.3 BoolQ Performance and Forgetting

The decline in BoolQ performance following fine-tuning provides clear evidence of catastrophic forgetting. Because the BoolQ task requires the model to interpret a passage and assess whether the passage supports a yes-or-no answer to a given question, its successful execution depends heavily on comprehension and reasoning skills. The drop from 57.80 percent accuracy to 50.40 percent accuracy suggests that the fine-tuning process altered internal representations that were previously useful for identifying textual entailment relationships. The fact that the final accuracy is almost indistinguishable from random chance indicates that the model's ability to perform passage-grounded reasoning degraded significantly. This degradation highlights the sensitivity of compact language models to narrow fine-tuning and illustrates how task-specific optimization can negatively affect general-purpose reasoning.

## 5.4 Sample Generations

Qualitative examples provide additional insight into how the fine-tuned model's behavior differs from the baseline model. When presented with an overtly positive text such as "I absolutely love this product! It's amazing," the baseline model responded with a generic phrase like "I love this," whereas the fine-tuned model produced the correct sentiment label "positive." A similar pattern appeared with neutral and negative examples. For instance, when given the sentence "The weather today is okay, nothing special," the baseline model produced a vague continuation, but the fine-tuned model correctly generated the label "neutral." In the case of strongly negative input such as "This is the worst experience I've ever had," the baseline model again generated a generic continuation, while the fine-tuned model responded with "negative." These examples show that the fine-tuned model internalized the instruction-style classification format and learned to respond with direct sentiment labels, whereas the baseline model had no inherent expectation to generate labels rather than natural text continuations. This contrast highlights the success of the fine-tuning process on the target task, even as it caused degradation on the secondary task.

# 6: Discussion and Conclusion

## 6.1 Interpretation of Results

The results of this project reveal a clear pattern: DistilGPT-2 is highly responsive to task-specific fine-tuning, but this adaptability comes at the cost of losing previously acquired capabilities. The improvement in sentiment classification accuracy demonstrates that the instruction-style fine-tuning process successfully guided the model to recognize emotional patterns and generate appropriate sentiment labels. The structure of the prompts and the repetitions within the dataset allowed the model to internalize the classification format and distinguish among negative, neutral, and positive tones with much greater confidence. In contrast, the significant decline in BoolQ accuracy shows that the model's ability to perform passage-grounded reasoning deteriorated after fine-tuning. This outcome suggests that the fine-tuning process overwrote or reshaped internal representations needed for the BoolQ task. Since the two tasks require distinct forms of understanding—emotional interpretation versus objective reasoning—the shift toward sentiment classification diminished the model's prior balance of capabilities. The findings align with prior research on catastrophic forgetting and underscore the difficulty of preserving general-purpose reasoning skills in models with limited capacity.

## 6.2 Trade-off Analysis

The interplay between performance gains on the fine-tuned task and losses on the secondary task demonstrates the presence of a measurable trade-off. The model achieved a substantial improvement of thirteen percentage points on sentiment classification but experienced a loss of more than seven percentage points on BoolQ. When interpreted proportionally, the model gained approximately one and three-fourths percentage points on sentiment accuracy for every percentage point it lost on BoolQ performance. This ratio provides a tangible way to understand the costs and benefits of fine-tuning a compact model such as DistilGPT-2. Although the improvement in the target task may seem desirable, the accompanying loss in general reasoning suggests that fine-tuning must be carefully managed when the preservation of broader capabilities is important.

### 6.3 Limitations

Several factors limit the generalizability of the results. The sentiment evaluation used only one hundred test samples for detailed confusion matrix analysis, which may not fully capture the range of edge cases present in the larger dataset. The project evaluated catastrophic forgetting using only one secondary task, BoolQ, whereas additional tasks could have provided a more comprehensive view of the model's retained or lost capabilities. The model itself is relatively small, with eighty-two million parameters, which likely contributes to its sensitivity to narrow fine-tuning. Larger models might be more robust to forgetting. Finally, the training process involved only two epochs of fine-tuning; different training durations or alternative learning rate schedules might produce different forgetting dynamics.

### 6.4 Conclusion

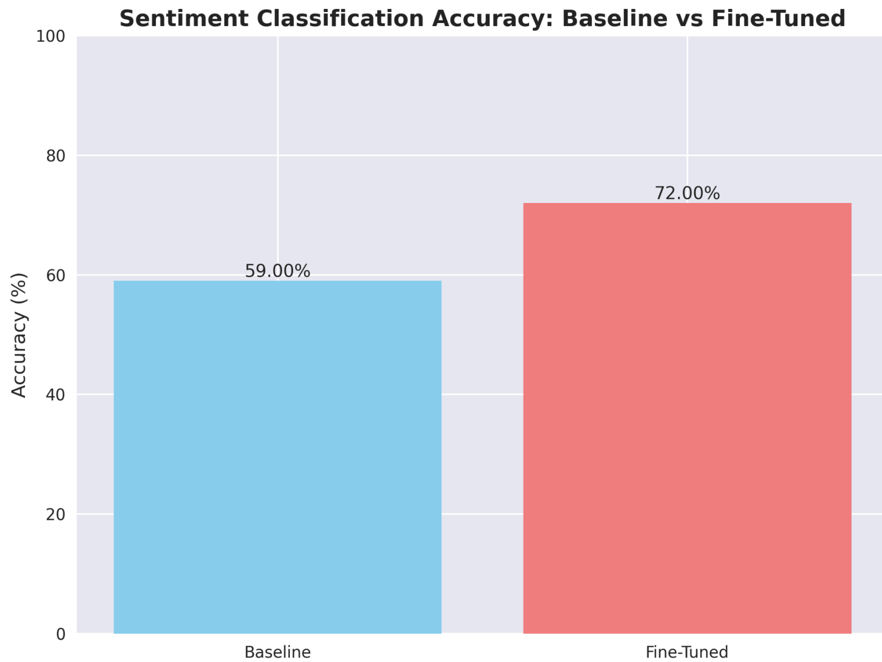
This project demonstrates that fine-tuning DistilGPT-2 on a specialized sentiment classification task leads to meaningful improvements on the target task while causing substantial degradation on an unrelated Boolean question answering task. The improvement in sentiment accuracy indicates that the model effectively learned the instruction-style classification format, but the drop in BoolQ accuracy highlights the vulnerability of compact models to catastrophic forgetting. These results reinforce the importance of understanding and managing the trade-offs associated with fine-tuning transformer-based language models, particularly in settings where maintaining broad generalization capabilities is essential.

### 6.5 Future Work

Future extensions of this project could focus on mitigation strategies designed to preserve previously learned capabilities while still enabling effective fine-tuning. Techniques such as Elastic Weight Consolidation could help prevent important weights from drifting too far during training. Parameter-efficient fine-tuning methods like LoRA or adapter-based approaches could reduce the degree to which the core model weights are modified, thereby limiting forgetting. Multi-task learning setups could allow the model to maintain proficiency across both sentiment classification and BoolQ. Additional evaluations involving multiple secondary tasks, including natural language inference and reading comprehension benchmarks, could also provide a clearer picture of the breadth of knowledge lost during fine-tuning. By investigating these directions, future work can help illuminate practical approaches for deploying fine-tuned models without sacrificing general reasoning capabilities.

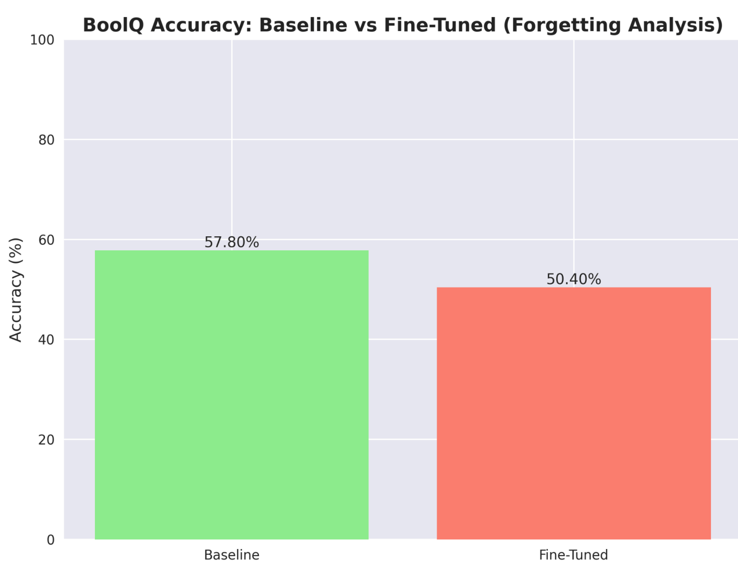
## 7: Figures and Visualizations

## Figure 1: Sentiment Classification Accuracy Comparison



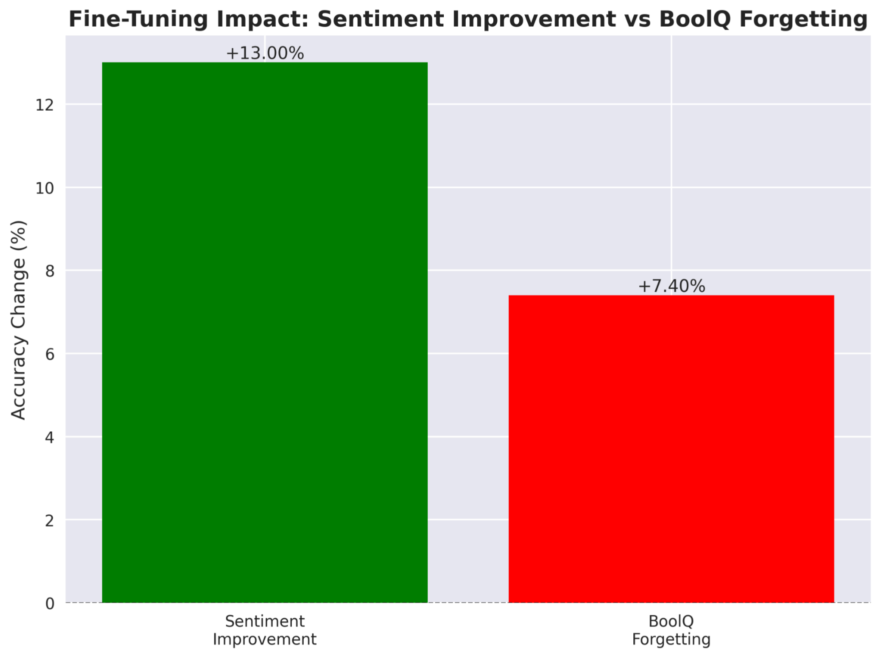

The first figure presents the bar chart comparing sentiment classification accuracy before and after fine-tuning. It shows a clear improvement, with accuracy increasing from 59.00 percent to 72.00 percent.

## Figure 2: BoolQ Accuracy Comparison



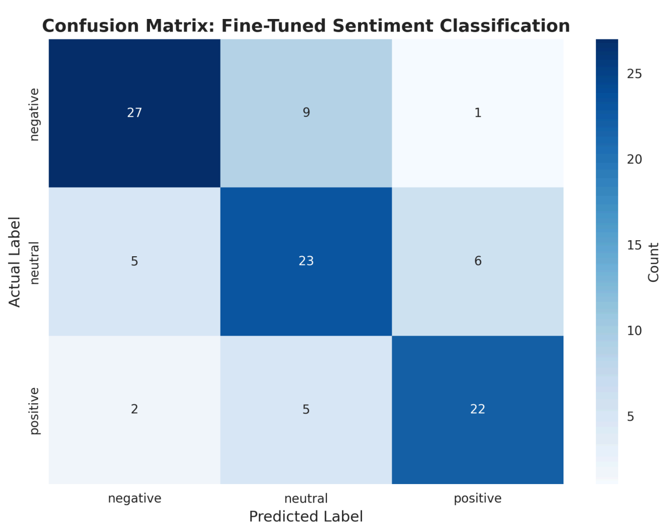

The second figure illustrates the drop in BoolQ accuracy following fine-tuning. The decline from 57.80 percent to 50.40 percent highlights the presence of catastrophic forgetting.

## Figure 3: Fine-Tuning Impact Summary




The third figure summarizes the overall trade-off between the performance gain on sentiment classification and the performance loss on BoolQ. It visually contrasts the improvement on the target task with the degradation on the secondary task.

## Figure 4: Sentiment Classification Confusion Matrix




The fourth figure displays the confusion matrix for the fine-tuned sentiment classifier, showing strong diagonal performance across the three sentiment classes. Most errors occur between adjacent categories, indicating that the model mainly struggles with borderline cases rather than extreme misclassifications.